MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 08 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | $(4 \times 1 = 4$ Marks)

Q1. If the system $kx + 3y = k - 3$ and $12x + ky = k$ has infinitely many solutions, then k is: [1]

(A) 6 (B) -6 (C) 0 (D) 12

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 6** [0.5 Mark]

Justification: $\frac{k}{12} = \frac{3}{k} = \frac{k-3}{k} \implies k^2 = 36 \implies k = \pm 6$, and $3 = k - 3 \implies \mathbf{k = 6}$. [0.5 Mark]

Q2. There are 20 cars and motorcycles in a parking area. If there are 56 wheels altogether, how many cars are there? [1]

(A) 12 (B) 10 (C) 8 (D) 14

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 8** [0.5 Mark]

Justification: Let cars $= x$, motorcycles $= y$. $x + y = 20$ and $4x + 2y = 56 \implies 2x + y = 28$. Subtracting: $x = \mathbf{8}$. [0.5 Mark]

Q3. The sum of two numbers is 20 and their product is 75. The sum of their reciprocals is: [1]

(A) 1/15 (B) 4/15 (C) 3/10 (D) 2/5

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 4/15** [0.5 Mark]

Justification: $\frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy} = \frac{20}{75} = \mathbf{\frac{4}{15}}$. [0.5 Mark]

Q4. The lines represented by $x - 4y + 14 = 0$ and $3x + 2y - 14 = 0$ intersect at the point: [1]

(A) $(2, 4)$ (B) $(-2, 4)$ (C) $(4, 2)$ (D) $(2, -4)$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) $(2, 4)$** [0.5 Mark]

Justification: Adding Eq. 1 and $2\times$ Eq. 2: $(x - 4y + 14) + (6x + 4y - 28) = 0 \implies 7x = 14 \implies x = 2 \implies y = 4$. [0.5 Mark]

SECTION B: Assertion-Reason Questions — MATHS MANTHAN | (2 × 1 = 2 Marks)

Q5. **Assertion (A):** The pair of equations $x+2y-5=0$ and $2x+4y-10=0$ has infinitely many solutions.
Reason (R): If $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$, then the linear equations represent coincident lines. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: $\frac{1}{2}=\frac{2}{4}=\frac{-5}{-10}=\frac{1}{2}$, so the lines coincide and share infinitely many solutions. Both true and (R) explains (A). [0.5 Mark]

Q6. **Assertion (A):** The point $(0,4)$ lies on the straight line $3x+2y=8$.
Reason (R): Substituting $x=0$ and $y=4$ into $3x+2y$ yields $3(0)+2(4)=8$. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: Because substituting $(0,4)$ gives $3(0)+2(4)=8=$ RHS, the point satisfies the equation and lies on the line. [0.5 Mark]

SECTION C: Very Short Answer Questions — MATHS MANTHAN | (3 × 2 = 6 Marks)

Q7. Find the value of k for which the lines $5x-7y=3$ and $15x+ky=9$ represent coincident lines. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Ratio Condition for Coincident Lines [1 Mark]

$\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2} \implies \frac{5}{15}=\frac{-7}{k}=\frac{3}{9}$.

Step 2: Solving for k [1 Mark]

$\frac{1}{3}=\frac{-7}{k} \implies \boxed{\mathbf{k=-21}}$.

Q8. Solve for x and y: $0.2x+0.3y=1.3$ and $0.4x+0.5y=2.3$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Removing Decimals ($\times 10$) [1 Mark]

$2x+3y=13$ — (i) and $4x+5y=23$ — (ii)

Step 2: Elimination [1 Mark]

Multiplying (i) by 2: $4x+6y=26$ — (iii). Subtracting (ii) from (iii): $y=3$.

From (i): $2x+3(3)=13 \implies 2x=4 \implies x=2$. $\boxed{\mathbf{x=2,\quad y=3}}$

Q9. For what value of p does the pair $4x+py+8=0$ and $2x+2y+2=0$ have a unique solution? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Formula / Theorem Used: Unique Solution Condition: $\frac{a_1}{a_2}\neq\frac{b_1}{b_2}$

Step 1 & 2: Ratio Comparison [2 Marks]

$\frac{4}{2}\neq\frac{p}{2} \implies 2\neq\frac{p}{2} \implies \boxed{\mathbf{p\neq 4}}$ (i.e., all real values of p except 4).

Q10. Solve for x and y: $a(x+y)+b(x-y)=a^2-ab+b^2$ and $a(x+y)-b(x-y)=a^2+ab+b^2$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Solving for $(x+y)$ and $(x-y)$ via Add-Subtract [1.5 Marks]

Let $a(x+y)+b(x-y)=a^2-ab+b^2$ — (i) $\quad a(x+y)-b(x-y)=a^2+ab+b^2$ — (ii)

Adding (i) and (ii): $2a(x+y)=2(a^2+b^2) \implies x+y=\dfrac{a^2+b^2}{a}$ $(a\neq 0)$ — (iii)

Subtracting (ii) from (i): $2b(x-y)=-2ab \implies x-y=-a$ $(b\neq 0)$ — (iv)

Step 2: Solving (iii) and (iv) for x and y [1.5 Marks]

Adding (iii) and (iv): $2x=\dfrac{a^2+b^2}{a}-a=\dfrac{a^2+b^2-a^2}{a}=\dfrac{b^2}{a} \implies x=\dfrac{b^2}{2a}$.

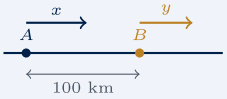
From (iv): $y=x+a=\dfrac{b^2}{2a}+a=\dfrac{2a^2+b^2}{2a}$.

$$\boxed{\mathbf{x=\frac{b^2}{2a},\qquad y=\frac{2a^2+b^2}{2a}}}$$

Q11. Places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If the cars travel in the same direction at different speeds, they meet in 5 hours. If they travel towards each other, they meet in 1 hour. What are the speeds of the two cars? [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Speed of Car from $A = x$ km/h, Car from $B = y$ km/h $(x > y)$.



Formula / Theorem
Used: Same Dir: $5(x-y)=100$; Opposite: $1(x+y)=100$

Step 1: Relative Speed Equations [1.5 Marks]

Same direction: $x-y=\frac{100}{5}=20$ — (i)

Opposite direction: $x+y=\frac{100}{1}=100$ — (ii)

Step 2: Solving for x and y [1.5 Marks]

Adding (i) and (ii): $2x=120 \implies x=60$ km/h. From (ii): $y=100-60=40$ km/h.

Speeds: **60** km/h and **40** km/h

Q12. The area of a rectangle gets reduced by 9 sq. units, if its length is reduced by 5 units and breadth is increased by 3 units. If we increase the length by 3 units and the breadth by 2 units, the area increases by 67 sq. units. Find the dimensions of the rectangle. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let length $= x$ units, breadth $= y$ units (Area $= xy$).

Step 1: Framing Equations [1.5 Marks]

$(x-5)(y+3)=xy-9 \implies xy+3x-5y-15=xy-9 \implies 3x-5y=6$ — (i)

$(x+3)(y+2)=xy+67 \implies xy+2x+3y+6=xy+67 \implies 2x+3y=61$ — (ii)

Step 2: Elimination [1.5 Marks]

Multiplying (i) by 3 and (ii) by 5: $9x-15y=18$ and $10x+15y=305$.

Adding: $19x=323 \implies x=17$ units. From (i): $3(17)-5y=6 \implies 5y=45 \implies y=9$ units.

Length = **17** units, Breadth = **9** units

Q13. The sum of a two-digit number and the number formed by reversing the digits is 132. If 12 is added to the number, the new number becomes 5 times the sum of the digits. Find the original number. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let tens digit $= x$ and units digit $= y$. Number $= 10x + y$, Reversed $= 10y + x$.

Step 1: Translating Both Conditions [2 Marks]

$(10x + y) + (10y + x) = 132 \implies 11(x + y) = 132 \implies x + y = 12$ — (i)

$(10x + y) + 12 = 5(x + y) \implies 10x + y + 12 = 5x + 5y \implies 5x - 4y = -12$ — (ii)

Step 2: Solving for x and y [2 Marks]

Multiplying (i) by 4: $4x + 4y = 48$ — (iii). Adding (ii) and (iii): $9x = 36 \implies x = 4$.

From (i): $4 + y = 12 \implies y = 8$.

Step 3: Constructing Original Number [1 Mark]

Original number $= 10(4) + 8 = \boxed{48}$ (Check: $48 + 84 = 132$, $48 + 12 = 60 = 5 \times 12$).

Q14. CASE STUDY: Multilevel Automated Parking Vehicle Toll Matrix

At a modern multilevel automated parking complex near a metro station, electronic sensors record vehicles entering the basement. The complex accommodates two-wheelers (scooters/motorcycles) and four-wheelers (cars). During the morning peak hour, the entrance boom barrier registered 150 vehicles entering the lot. The underground pressure sensors recorded a total of 420 rolling wheels. Let x denote the number of two-wheelers and y denote the number of four-wheelers. [4]

(i) Formulate the pair of linear equations in x and y modeling the vehicle count and wheel count. [1]

(ii) Determine the total number of two-wheelers parked in the facility (x). [1]

(iii) Determine the total number of four-wheelers parked in the facility (y). [1]

(iv) If the parking fee is Rs. 20 per two-wheeler and Rs. 50 per car, calculate the total revenue generated during the morning rush. [1]

— OR —

Find the difference between the revenue collected from cars and two-wheelers. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Formula / Theorem Used: Total Vehicles $= x + y$, Total Wheels $= 2x + 4y$

Sub-part (i): Modeling Equations [1 Mark]

$x + y = 150$ — (i) and $2x + 4y = 420$ (or $x + 2y = 210$) — (ii)

Sub-part (ii): Number of Two-Wheelers (x) [1 Mark]

Subtracting (i) from $x + 2y = 210$: $y = 210 - 150 = 60$. Hence $x = 150 - 60 =$ **90** two-wheelers.

Sub-part (iii): Number of Four-Wheelers (y) [1 Mark]

From the elimination step above: **60** four-wheelers.

Sub-part (iv): Total Revenue Collected [1 Mark]

Revenue $= 20(90) + 50(60) = 1800 + 3000 =$ Rs. **4,800**.

Sub-part (iv) [OR Option]: Difference Between Car & Two-Wheeler Revenue [1 Mark]

Difference $= 3000 - 1800 =$ Rs. **1,200**.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★